VIJAY MALVIYA

Backend-Focused Full-Stack Engineer

+91-98934-97541 | ashum3486@gmail.com | Portfolio/Vijay-Malviya | GitHub: github.com/Vijay9424

PROFESSIONAL SUMMARY

Backend-focused full-stack engineer with expertise in building scalable, real-time distributed systems. Strong foundation in Node.js, Express, MongoDB, Redis, and Next.js. Experienced architecting real-time communication features using Socket.IO and MQTT, with focus on low-latency API performance and system optimization. Currently developing a social commerce platform emphasizing modular microservice architecture and enterprise-grade backend systems.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Node.js, Bash
Frontend: React, Next.js (App Router), Tailwind CSS, Framer Motion, Redux, Context API
Backend: Express.js, Flask, Django, REST APIs, WebSocket, Socket.IO, MQTT, JWT
Databases: MongoDB (indexing, cursor pagination), Redis (cache-aside, locks, TTL), Elasticsearch, Cloudinary
Infrastructure: Docker, GitHub Actions, Vercel, Render, Nginx
Architecture: Microservices, BullMQ workers, Brotli compression, HLS streaming, Database optimization
Security: JWT refresh token rotation, bcrypt, RSA encryption, Rate limiting, Input validation (Joi), Structured logging
ML & Data: TensorFlow, scikit-learn, OpenCV, NLP pipelines, Data preprocessing, Model deployment

PROFESSIONAL EXPERIENCE & PROJECTS

Full-Stack Social Commerce Platform (In Development) Jan 2025 – Present | Next.js Node.js MongoDB Socket.IO Redis

- Architecting scalable real-time social commerce platform integrating messaging, video calling, product listings, and personalized feeds using microservice-based backend design.
- Engineered Socket.IO real-time communication layer with Redis adapter for horizontal scalability; optimized delivery latency through connection pooling and message batching.
- Structured MongoDB schemas using compound indexes and lean query projections; enhanced feed performance through pagination and caching strategies.
- Established Redis cache-aside pattern with TTL-based key expiration; implemented distributed locking to prevent cache stampedes during high traffic.
- Developed secure JWT authentication with refresh token rotation and Redis-based session management for stateless API access control.
- Integrated Cloudinary for media storage; configured chunked uploads, CDN caching, and adaptive image transformations for optimal delivery.
- Enabled multi-environment deployments on Vercel (frontend) and Render (backend).
- Developed PWA features including offline support, background sync, and push notifications for cross-device user experience.

NLP Emotion Detection from Text Sep 2022 – Nov 2022 | Python TensorFlow scikit-learn Flask

- Trained deep neural networks on 10,000+ labeled sentences achieving 94% accuracy across five emotion classes (joy, sadness, anger, fear, neutral).
- Developed complete preprocessing pipeline: tokenization, lemmatization, GloVe word embeddings (100D), and SMOTE for handling class imbalance.
- Exposed trained model via Flask REST API; processed 1,000+ requests/hour with sub-100ms latency using TensorFlow Lite optimization.
- Optimized model inference for production deployment; implemented batch prediction for bulk text analysis reducing computational overhead.

Deep Learning Personality Prediction via Handwriting Analysis Dec 2022 – Feb 2023 | Python OpenCV TensorFlow CNN

- Developed CNN-based handwriting personality classifier using custom dataset of 500 handwriting samples with diverse demographics.
- Applied preprocessing techniques including skew correction, contrast normalization, and stroke width optimization; expanded dataset 5x through data augmentation (rotation, scaling, elastic deformation).
- Trained ResNet-based architecture achieving 78% accuracy; evaluated model performance using k-fold cross-validation and confusion matrix analysis.
- Analyzed feature importance using Grad-CAM visualization to identify key handwriting characteristics influencing personality predictions.

BACKEND ARCHITECTURE & SYSTEM DESIGN EXPERTISE

- **Scalability & Performance:** Architected stateless microservices enabling horizontal scaling. Established cursor-based pagination eliminating skip/limit performance issues on large datasets. Achieved sub-50ms internal response times through query optimization and caching.
- **Caching & Optimization:** Orchestrated Redis cache-aside pattern with distributed locking preventing cache stampedes. Stored Brotli-compressed payloads as binary buffers; configured key namespacing and TTL strategies optimizing memory utilization.
- **Authentication & Security:** Implemented JWT access tokens (5-15 min TTL) + refresh tokens (7d TTL) with JTI UUID rotation. Maintained per-user session arrays and JTI blacklist in Redis; applied bcrypt with salting; enforced rate limiting on auth endpoints.
- **Asynchronous Processing:** Launched BullMQ background job queues handling media processing, HLS transcoding, and email delivery. Designed idempotent workers with exponential backoff retry logic and priority-based task execution.
- **Media Management:** Configured Cloudinary canonical storage with eager/lazy transformations. Established chunked video uploads with resume capability; automated webhook-triggered processing and background worker integration.
- **Real-Time Systems:** Architected Socket.IO with Redis adapter enabling cross-server message broadcasting. Developed room-based messaging for targeted notifications; integrated MQTT broker for lightweight pub/sub patterns.
- **Observability:** Configured structured JSON logging (Winston); monitored critical metrics: Redis latency, queue backlog, API response times. Set up alerts for performance thresholds.

EDUCATION

Bachelor of Technology Computer Science & Engineering
Indian Institute of Information Technology (IIIT) Vadodara International Campus Diu
Expected Graduation: 2026
Relevant Coursework: Advanced Data Structures, Algorithm Design, Database Systems, Computer Networks, Information Theory, Machine Learning, Software Engineering

CERTIFICATIONS & ACHIEVEMENTS

- **NVIDIA Deep Learning Institute** Fundamentals of Deep Learning (Completed: March 2024)

ADDITIONAL INFORMATION

Languages: Hindi (Native) English (Fluent)
Portfolio: Vijay's Portfolio Full-stack projects
Interests: System design, enterprise scalability, distributed systems, real-time architectures, high-performance computing
Availability: Full-time opportunities from 2026; open for internships and part-time projects